DEBRIS CAREERS — PROJECT REPORT & VIVA GUIDE

AI, Natural Language Processing (NLP) & Decision-Tree Career Consultant

1. PROJECT METADATA & EXECUTIVE SUMMARY

• Project Name: Debris Careers
• Domain: Artificial Intelligence, Natural Language Processing (NLP) & Information Retrieval (IR)
• Purpose: A deterministic, personalized career consulting platform designed for college students and graduates.
• Core Philosophy: Zero hallucination, zero paid API dependency, 100% offline local machine learning.
• Key Technology Stack: Python 3, Flask, scikit-learn (TF-IDF & Cosine Similarity), Pandas, SQLite3, HTML5/CSS3.
• Visual Architecture: Minimalist Monochrome Liquid Glass (Glassmorphism), Porsche Typography, Dynamic Themes.

2. MOTIVATION & PROBLEM STATEMENT

• Career confusion is a major challenge for school and university students facing rapid industry transformations.
• Traditional web forms are rigid, passive, and fail to provide personalized or explainable guidance.
• Generative AI / LLM chatbots often hallucinate non-existent degrees, fabricate university details, or cost money via APIs.
• Solution: Debris Careers pairs a stateful multi-turn conversational interface with a mathematical TF-IDF
vectorization engine to deliver reliable, explainable recommendations backed by verified institutional data.

3. ARCHITECTURAL OVERVIEW

• Tier 1 (Presentation): Soothing Liquid-Glass web interface with ambient harmonic particles and Light/Dark themes.
• Tier 2 (Application Server): Lightweight Flask controller managing sessions, chat routes, and database operations.
• Tier 3 (Inference Engine): Dual-mode system featuring direct NLP attribute extraction and a multi-turn profiler.
• Tier 4 (Data & Persistence): Relational SQLite3 database storing users, consultations, and message history.

MACHINE LEARNING & NLP RECOMMENDATION PIPELINE

5-Stage Mathematical Pipeline for Deterministic Recommendations

STAGE 1: NLP ATTRIBUTE EXTRACTION

- When a student inputs a natural statement (e.g., 'I like mathematics and computers. I want a high-paying job'):
- The text is normalized, tokenized, and evaluated across semantic lexicons (Interests, Skills, Career Values).
- Extracts explicit priorities: High Salary, Job Stability, Creative Freedom, or Social Impact.

STAGE 2: TF-IDF VECTORIZATION (TERM FREQUENCY - INVERSE DOCUMENT FREQUENCY)

- Machine learning models operate on numerical representations rather than raw strings.
- Uses scikit-learn's TfidfVectorizer with sublinear term-frequency scaling and unigram + bigram (1, 2) ranges.
- Formula: TF-IDF(t, d, D) = TF(t, d) * IDF(t, D), where IDF(t, D) = ln((1 + |D|) / (1 + DF(t, D))) + 1.
- Converts student input and all pre-curated career profiles into high-dimensional numerical feature vectors.

STAGE 3: COSINE SIMILARITY CALCULATION

- Computes the mathematical angle between the student's vector (S) and each career feature vector (C):
- Formula: Cosine_Similarity(S, C) = (S • C) / (||S|| * ||C||) = Sum(S_i * C_i) / (Sqrt(Sum(S_i^2)) * Sqrt(Sum(C_i^2))).
- Yields a normalized relevance score between 0.00 and 1.00 (displayed as a percentage confidence bar).

STAGE 4 & 5: RANKING & VERIFIED EXPLANATION SYNTHESIS

- Sorts cosine similarity scores in descending order and selects the Top 3 best matches.
- Explains WHY the match occurred based on extracted keywords, avoiding opaque black-box outputs.
- Delivers: Required Degrees, Top Tier Colleges (IITs, AIIMS, SRCC, NLSIU), Top Recruiters, and a 5-Step Action Plan.

CONVERSATIONAL FLOW & DATABASE SECURITY

Multi-Turn Dialogue Management & Relational SQLite Architecture

MULTI-TURN CHATBOT VS. STATIC FORMS

• Unlike a one-off form, Debris Careers conducts an interactive, multi-turn consultation:
  - Turn 1: Academic subjects and foundational interests (e.g., Math, Bio, Accounts, Law).
  - Turn 2: Day-to-day functional activities (e.g., Coding, Clinical Diagnosis, Auditing, Visual Design).
  - Turn 3: Core career drivers and priorities (e.g., High Salary, Stability, Creativity, Social Impact).
• Flexible Execution: Users can also input a complete natural prompt at any time to trigger instant scoring.

DATABASE SCHEMA & PERSISTENCE (SQLite3)

• Table 1: users (id, email UNIQUE, password_hash, created_at)
• Table 2: conversations (id UUID PRIMARY KEY, user_id FK, title, state_data JSON, created_at, updated_at)
• Table 3: messages (id, conversation_id FK, sender ['user'/'bot'], content, options JSON, timestamp)
• Full Session Retention: Chat histories persist across logins, with options to delete individual chats or clear all.

SECURITY CONSTRAINTS & PASSWORD POLICY

• Explicit Constraint: Maximum password length of 8 characters.
• Dual-Layer Enforcement: Enforced at both presentation (HTML maxlength='8') and server (Python len <= 8) layers.
• Cryptographic Hashing: Passwords are salted and hashed using SHA-256 before database insertion.

AESTHETICS, UI INNOVATIONS & VIVA CHEATSHEET

Design Psychology & Comprehensive Q&A for Evaluators

SOOTHING VISUAL AESTHETICS (GLASSMORPHISM & BEIGE PALETTE)

- Design Purpose: Reduces career planning anxiety and cognitive fatigue.
- Warm Frosted Beige Palette: Warm cream (#FAF6F0) and oat canvas with frosted liquid-glass panels.
- Porsche Headline Typography: Exclusive '911 Porscha' headline typography paired with modern sans-serif body text.
- Ambient Constellation Canvas: HTML5 canvas displaying 60-75 gently floating particles in harmonic sine-wave orbits.
- Dynamic Theme Toggle: Instant transition between soothing Warm Beige and deep Cosmic Obsidian Dark Mode.

COLLEGE VIVA CHEATSHEET — HIGH-YIELD QUESTIONS & ANSWERS

Q1: Why use TF-IDF and Cosine Similarity instead of OpenAI / ChatGPT API?
Ans: LLMs are prone to hallucinations, require paid API keys, and have network latency. Our approach runs 100% offline,
produces explainable mathematical similarity scores, and guarantees accurate college and recruiter information.

Q2: How does the chatbot recognize keywords without an LLM?
Ans: We use a dual-tier NLP architecture: Step 1 extracts semantic entities (subjects, priorities); Step 2 calculates
cosine similarity across n-gram vector spaces, mapping student intent to verified career profiles.

Q3: How are career recommendations explained to students?
Ans: Rather than generating text probabilistically, the system synthesizes verified templates detailing matching traits,
academic pathways, top colleges (e.g., IITs, SRCC, AIIMS), top recruiters (e.g., Google, Deloitte), and next steps.